

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-008 - Nahorkatiya-008

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000086
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Kopili (KOP)
Spud / Completion	2019-09-11 / 2020-02-02
Measured / TVD	3270 m / 3224 m
Trajectory	Deviated; inc 13.1 deg; az 47.4 deg
Formation Interval	1780-2729 m
Lithology	Calcareous shale and siltstone
Pore Pressure Reference	3524 psi
Fracture Gradient	15.18 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	High (59.1/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	817	Surface
9 5/8 in casing	1896	Intermediate
7 in liner/casing	2877	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2019-11-10 1517 m	Mud Loss	Critical	Intermittent circulation losses developed while drilling ahead. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Track flow balance and pit volume trend before losses become severe.
2019-11-12 1545 m	Mud Loss	Critical	Flow-out dropped below flow-in with pit loss. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2019-10-01 608 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Avoid prolonged static periods in high-drag interval.
2019-10-09 788 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2019-10-15 935 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2019-11-02 1335 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2019-11-28 1889 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.

Completion Summary

The synthetic well reached 3270 m MD in the Kopili interval. The NWIS knowledge record links 14 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.